

Final Customer Profiling Report

Cluster Evaluation & Customer Profiling

Module 07 — Machine Learning | Customer Personality Analysis Dataset

1. Objective

This report converts the four K-Means clusters developed in Module 06 into meaningful, business-actionable customer segments by analyzing demographic characteristics, spending behaviour, shopping channel preferences, engagement, and marketing campaign responsiveness — providing customer profiles that support strategic marketing and decision-making.

2. Cluster Overview

Cluster	Segment Name	Size	% of Customers
3	High-Value Customers	495	22.1%
1	Premium / Loyal Buyers	639	28.6%
0	Discount Seekers / Budget Customers	427	19.1%
2	New / Developing Customers	676	30.2%

Cluster Distribution

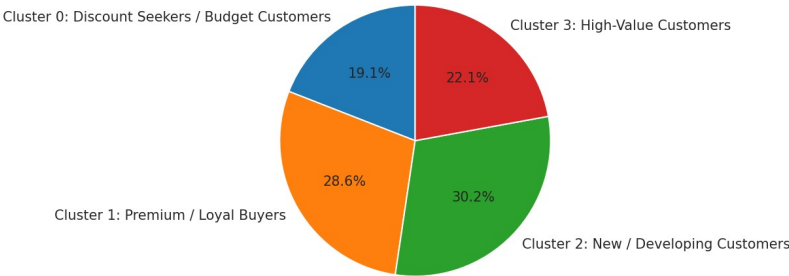


Figure 2.1 — Cluster size distribution.

3. Demographic Analysis (Activity 1)

Cluster	Segment	Avg Age	Avg Income	Family Size	Children
0	Discount Seekers / Budget	61.9	\$42,574	3.7	2.0
1	Premium / Loyal Buyers	60.8	\$59,927	2.8	1.1
2	New / Developing	50.5	\$31,891	2.4	0.8
3	High-Value	57.1	\$76,705	1.7	0.1

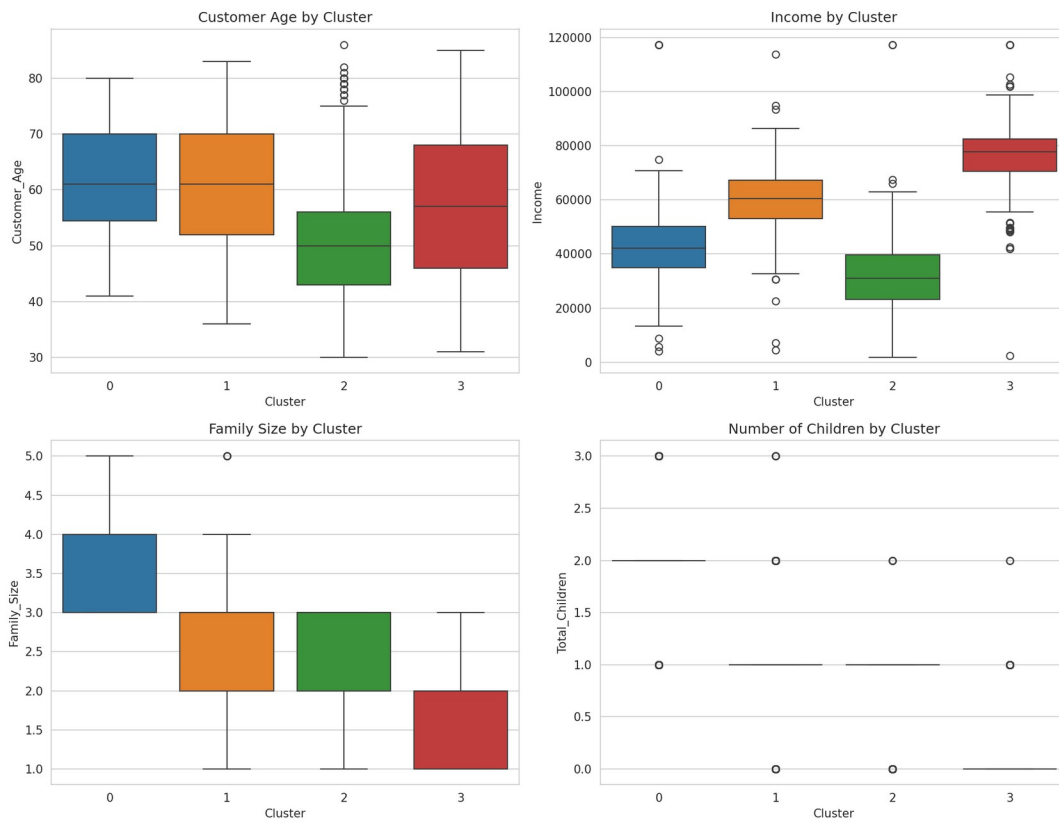


Figure 3.1 — Age, income, family size, and children by cluster.

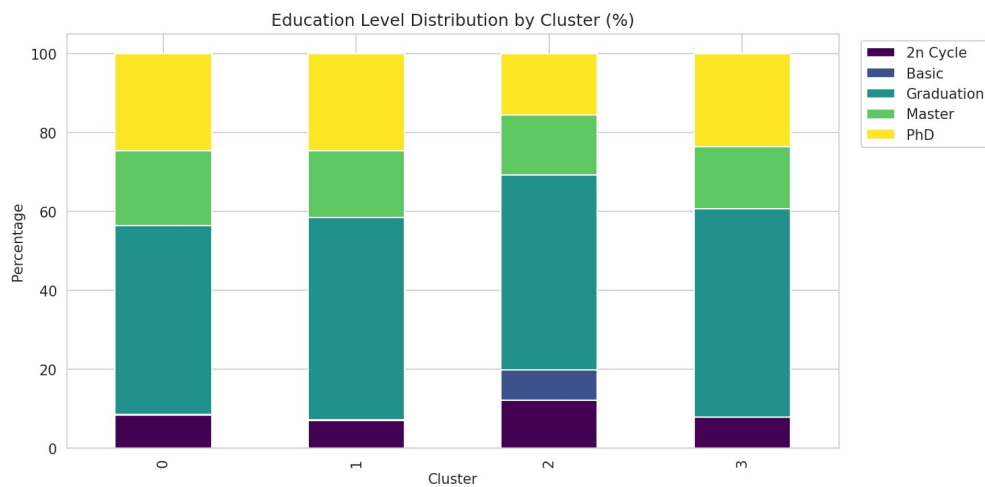


Figure 3.2 — Education level distribution by cluster.

4. Spending Behavior Analysis (Activity 2)

Cluster	Wine	Fruits	Meat	Fish	Sweets	Gold	Total
0	\$75	\$4	\$27	\$6	\$4	\$16	\$131
1	\$469	\$34	\$164	\$45	\$36	\$67	\$814
2	\$37	\$7	\$27	\$10	\$7	\$18	\$105
3	\$654	\$63	\$481	\$93	\$64	\$75	\$1,430

Cluster 3 (High-Value) is the highest spender across every category and the top premium (Wine + Gold) buyer. Cluster 2 (New/Developing) is the lowest spender overall — the budget-conscious segment.

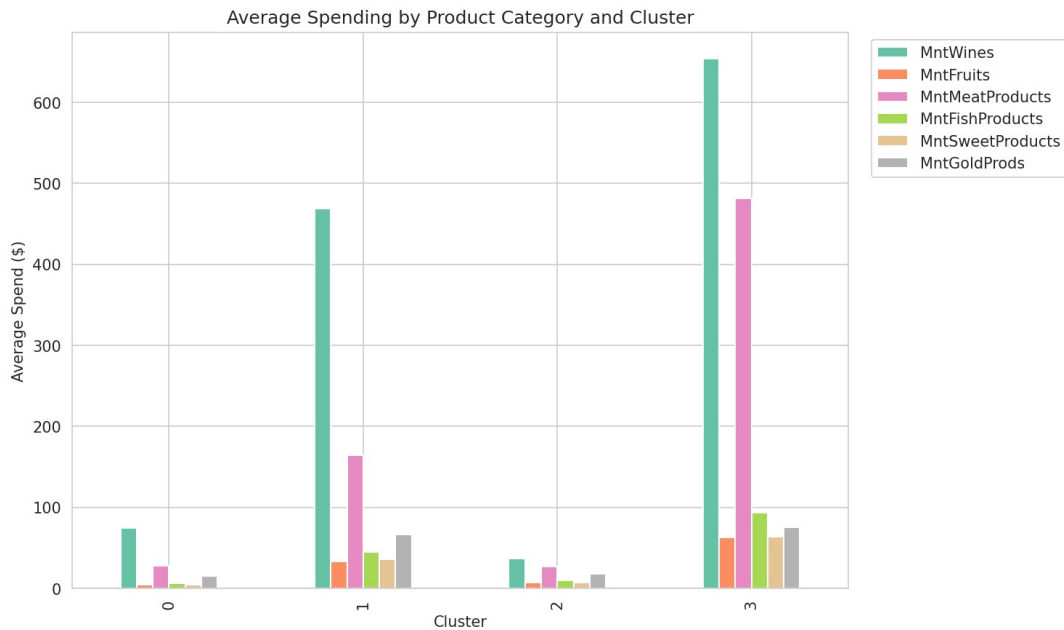


Figure 4.1 — Average spending by product category and cluster.

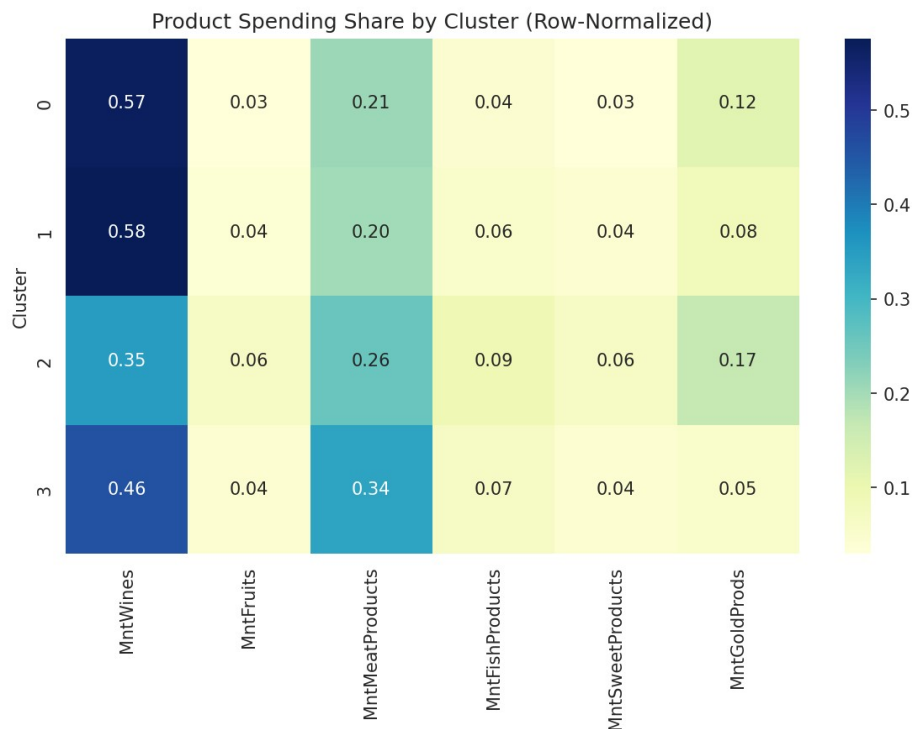


Figure 4.2 — Product spending share by cluster (normalized).

5. Shopping Channel & Engagement Analysis (Activity 3)

Cluster	Web	Catalog	Store	Deals	Web Visits/mo	Recency (days)
0	2.4	0.8	3.5	2.8	6.2	50.2
1	6.7	3.4	8.2	3.5	5.5	47.7

Cluster	Web	Catalog	Store	Deals	Web Visits/mo	Recency (days)
2	2.1	0.6	3.2	1.8	6.5	49.0
3	4.8	6.3	8.2	1.2	2.7	50.1

- Digital-First / Store-Oriented / Deal-Seeking: Cluster 1 (Premium/Loyal Buyers) leads on web, store, and deal purchases.
- Catalog-Oriented: Cluster 3 (High-Value) makes the most catalogue purchases.
- Recency is nearly flat across all clusters (47.7–50.2 days) — it is not a strong differentiator in this dataset.

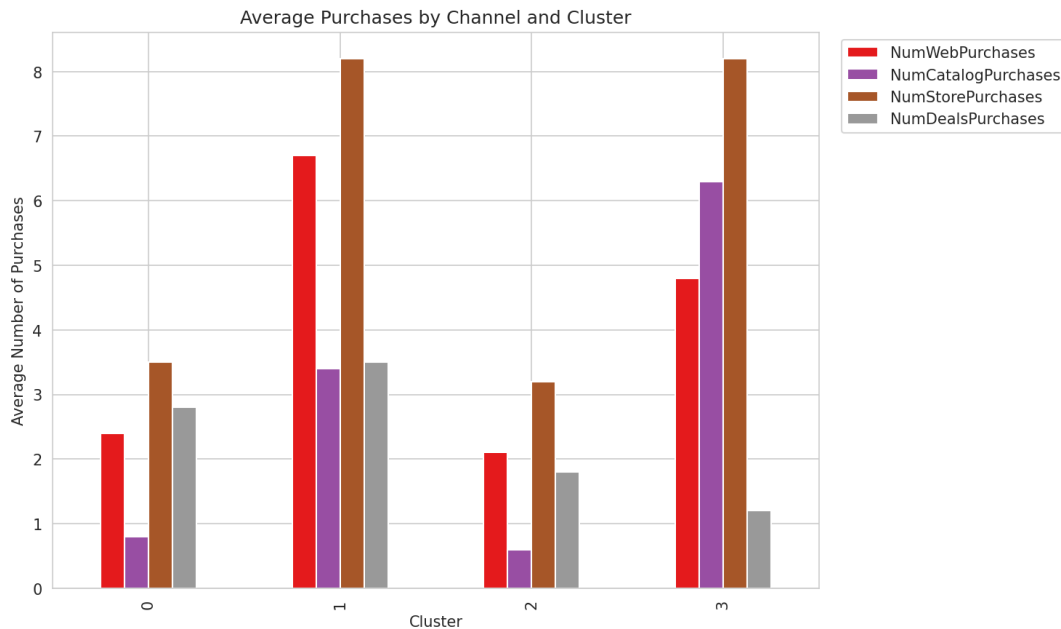


Figure 5.1 — Average purchases by channel and cluster.

6. Marketing Campaign Analysis (Activity 4)

Cluster	Cmp1	Cmp2	Cmp3	Cmp4	Cmp5	Latest	Complaints
0	0.9%	0.5%	6.1%	2.8%	0.0%	7.5%	1.2%
1	2.5%	0.6%	5.9%	10.2%	1.4%	9.7%	1.1%
2	0.1%	0.0%	7.4%	1.0%	0.0%	9.9%	0.9%
3	24.8%	4.8%	9.9%	16.8%	30.9%	34.9%	0.4%

Cluster 3 (High-Value) is by far the most campaign-responsive segment across every campaign, especially Campaign 5 (30.9%) and the Latest Campaign (34.9%). Cluster 0 (Discount Seekers) is the most marketing-resistant segment and — combined with its comparatively higher Recency — is the priority target for re-engagement.

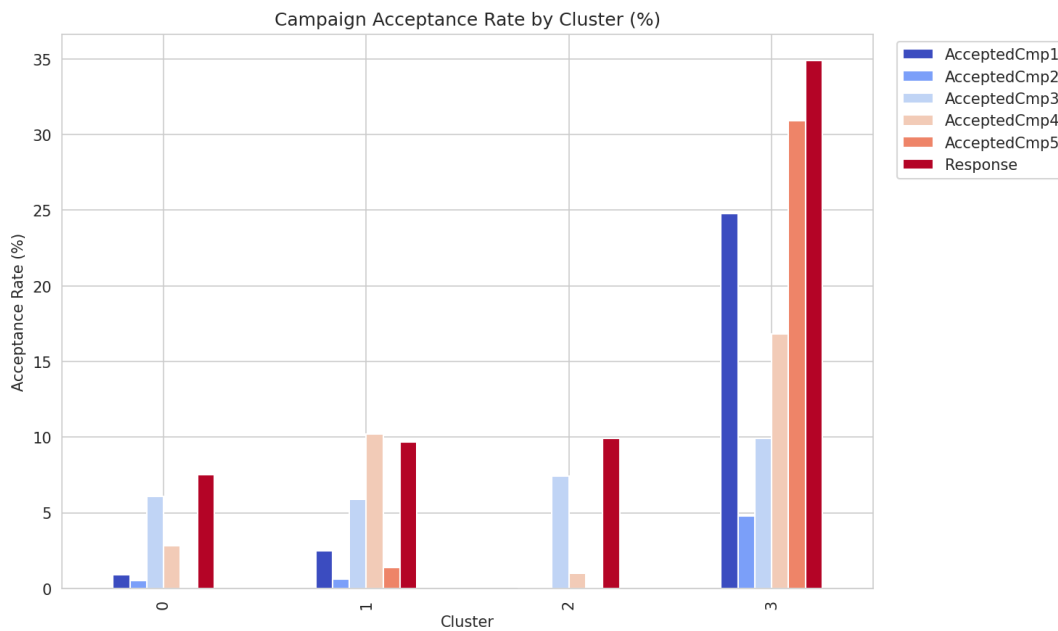


Figure 6.1 — Campaign acceptance rate by cluster.

7. Business Segmentation & Cluster Naming (Activity 5)

Cluster	Business Name	Key Justification
3	High-Value Customers	Highest income, spend, and campaign acceptance; smallest households.
1	Premium / Loyal Buyers	Solid income and spend; most active across web, store, and catalogue.
0	Discount Seekers / Budget Customers	Lower income and spend, largest families, lowest campaign response.
2	New / Developing Customers	Lowest income and spend, youngest average age; growth potential segment.

8. Customer Persona Development (Activity 6)

Persona: Victoria the Premium Patron (Cluster 3 — High-Value Customers)

Age Range	49–65 yrs (avg 57)
Income Level	\$76,705/year
Family Status	Small household, avg family size 1.7
Shopping Habits	High-frequency buyer across catalogue and store; ~73 purchases per period
Preferred Product Category	Wine, Meat
Preferred Purchasing Channel	Store, Catalog
Marketing Responsiveness	34.9% latest-campaign acceptance — highly responsive
Customer Challenges	Risk of campaign fatigue from over-targeting; needs relevance over frequency
Recommended Marketing Strategy	Premium bundles, early access, VIP loyalty perks
Estimated Customer	~\$4,290 (3-year proxy)

Lifetime Value	
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Persona: Marcus the Multi-Channel Regular (Cluster 1 — Premium / Loyal Buyers)

Age Range	52–68 yrs (avg 61)
Income Level	\$59,927/year
Family Status	Avg family size 2.8
Shopping Habits	Most active shopper overall — leads in web, store, and deal purchases
Preferred Product Category	Wine, Meat
Preferred Purchasing Channel	Store, Web
Marketing Responsiveness	9.7% latest-campaign acceptance — moderate
Customer Challenges	Under-targeted by past campaigns relative to activity level
Recommended Marketing Strategy	Cross-channel cross-sell and loyalty rewards
Estimated Customer Lifetime Value	~\$2,442 (3-year proxy)

Persona: The Rodriguez Family — Budget-Conscious Household (Cluster 0 — Discount Seekers / Budget Customers)

Age Range	53–69 yrs (avg 62)
Income Level	\$42,574/year
Family Status	Largest households, avg family size 3.7
Shopping Habits	Deal-driven, moderate store purchases, low catalogue use
Preferred Product Category	Wine (budget-tier), Meat
Preferred Purchasing Channel	Store
Marketing Responsiveness	7.5% latest-campaign acceptance — lowest of all segments
Customer Challenges	Most marketing-resistant segment; standard campaigns underperform
Recommended Marketing Strategy	Value bundles and family-size discount promotions
Estimated Customer Lifetime Value	~\$394 (3-year proxy)

Persona: Jamie the New Explorer (Cluster 2 — New / Developing Customers)

Age Range	42–58 yrs (avg 50, youngest segment)
Income Level	\$31,891/year
Family Status	Avg family size 2.4
Shopping Habits	Lowest spend and purchase volume; still forming habits
Preferred Product Category	Wine (entry-level)

Preferred Purchasing Channel	Store
Marketing Responsiveness	9.9% latest-campaign acceptance — moderate
Customer Challenges	Low engagement and spend; largest segment by size (30.2%) with growth potential
Recommended Marketing Strategy	Onboarding journeys, welcome incentives, educational content
Estimated Customer Lifetime Value	~\$315 (3-year proxy, expected to grow with tenure)

9. Cluster Visualization Package (Activity 7)

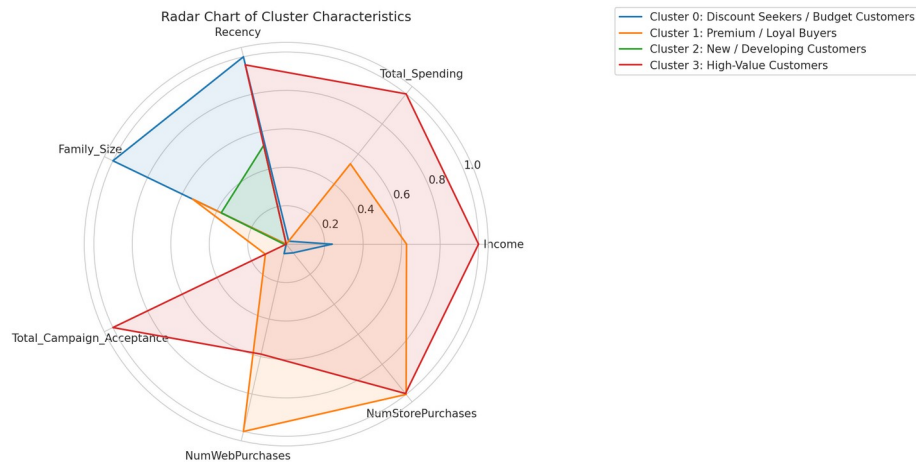


Figure 9.1 — Radar chart of normalized cluster characteristics.

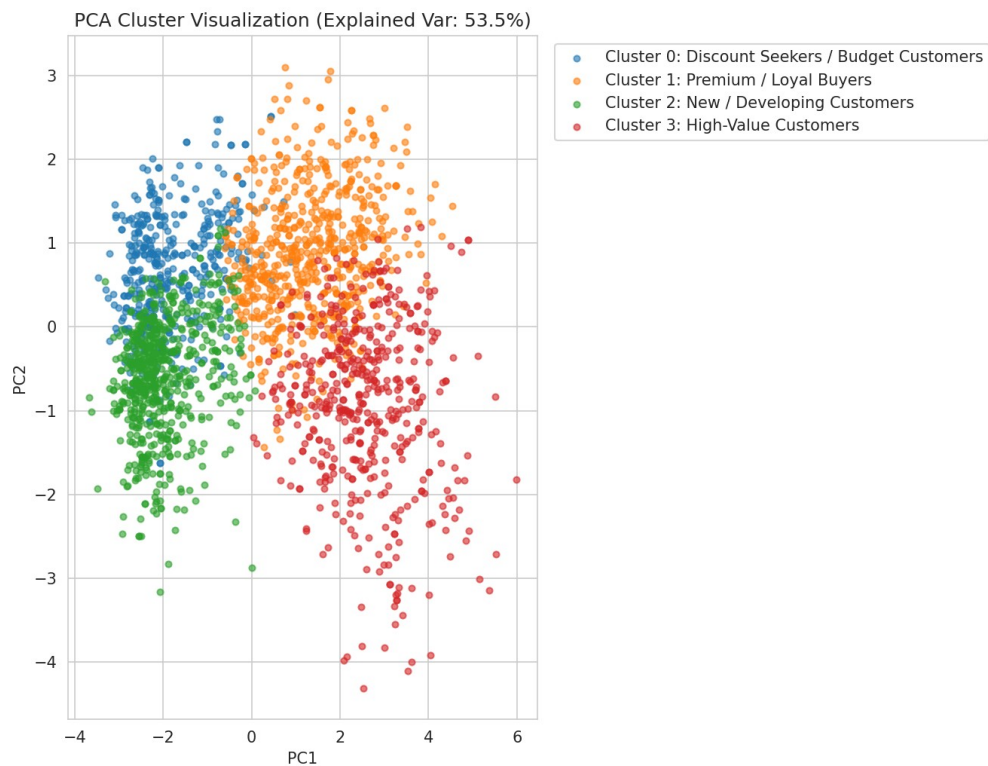


Figure 9.2 — PCA projection of clusters using 11 behavioural/demographic features.

10. Business Recommendations Summary

- High-Value Customers (22.1%): protect and grow this high-margin, highly responsive segment with premium/VIP treatment.
- Premium/Loyal Buyers (28.6%): the most active shoppers — increase campaign targeting to match their engagement level.
- Discount Seekers/Budget Customers (19.1%): the priority re-engagement segment — least responsive, largest families; needs value-led offers, not premium campaigns.
- New/Developing Customers (30.2%): largest segment by count — invest in onboarding to convert this group into higher-value buyers over time.